

Detecting Incomplete Gentrification: A Spline-Based Approach to Measuring Neighborhood Transformation Dynamics

Matteus Schmidt

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1 Introduction

1.1 Abstract

The quantification of incomplete gentrification poses a persistent challenge for urban researchers. Traditional typologies often fail to capture the early, uneven, or partial nature of these transformations, leaving gaps in how urban change is measured and understood. This study introduces a novel index to measure incomplete gentrification across U.S. ZIP codes from 2019 to 2023, offering a scalable and replicable approach to detect early-stage urban transformation. The metric serves as a proxy for the velocity of incomplete gentrification, while the specific composition of the index in a given ZIP code reveals the type and character of incomplete gentrification taking place.

1.2 Literature Review

Gentrification manifests differently across urban contexts, creating definitional challenges for researchers and policymakers. Most scholars characterize it as neighborhood socioeconomic transformation, typically involving rising average incomes (Lee, 2010) or accelerated changes in socioeconomic indicators relative to citywide averages (Grube-Cavers and Patterson, 2015). An important distinction raised by Freeman (2005) is whether neighborhood change represents "displacement or succession" - a crucial consideration for policy responses that must address both immediate displacement concerns and longer-term demographic transitions.

Chapple and Zuk (2016) emphasize the importance of early warning systems to identify neighborhoods experiencing incipient gentrification before full displacement occurs. Their work on neighborhood early warning systems underscores the need for proactive policy intervention during early transformation stages. This is particularly relevant for addressing incomplete gentrification, where neighborhood change has begun but hasn't yet resulted in wholesale displacement.

Precise gentrification metrics provide multiple benefits for urban research and policy development. Quantification helps identify driving factors behind neighborhood change, measure transformation intensity, and establish benchmarks for comparing affected areas. Evidence-based measurements are particularly valuable for policymakers developing targeted interventions to address displacement and preserve community stability, allowing them to allocate resources where they're most needed rather than relying on anecdotal evidence.

Spline-based models offer significant advantages for detecting subtle temporal patterns in gentrification processes. Unlike simple linear trends, splines can capture both the direction and acceleration of neighborhood change, revealing early signals of transformation that might otherwise remain undetected. This mathematical approach provides a more nuanced representation of how socio-economic indicators evolve over time, pinpointing the driving force behind the analysis of Bunten et. al.

Building on Bunten's framework for re-measuring gentrification, spline models extend our analytical capacity by quantifying not just if change is occurring, but how that change is unfolding over time. Bunten's work emphasizes the importance of detecting differential rates of change across neighborhoods—a perspective that aligns perfectly with spline-based approaches that can distinguish between linear, accelerating, and decelerating patterns of transformation. By incorporating the curvature parameters from quadratic splines, researchers can identify neighborhoods where gentrification pressures are intensifying (positive curvature), stabilizing (near-zero curvature), or potentially reversing (negative curvature).

Furthermore, this methodological approach addresses a key limitation in traditional gentrification metrics by focusing on the velocity and acceleration of change rather than absolute thresholds. When combined with principal component analysis of demographic shifts, as demonstrated in studies of incomplete gentrification, spline-derived indicators create a more dynamic and responsive measurement framework. This allows for the identification of neighborhoods experiencing gentrification pressures at various stages—from early warning signals to advanced transformation—providing policymakers with critical lead time to implement preventative measures before widespread displacement occurs.

2 Methodologies

2.1 Overview

The index synthesizes temporal trends in six key indicators: median income, educational attainment, homeownership, rent burden, poverty rate, and displacement. These metrics are represented by a summation of their slopes and curvatures to create a composite measure of change using spline-based models. The final metric is constructed via principal component analysis on racial and ethnic composition changes by zip to capture demographic turnover, particularly reflecting minority displacement patterns by non-hispanic whites. These features are integrated with a the vulnerability index adapted from the Urban Displacement Project to reflect structural susceptibility to gentrification pressures, and identify stages of incomplete gentrification.

All data was based on the Annual Community Survey conducted by the Census bureau. Non-metropolitan ZIP codes were excluded using RUCC classifications to focus on urban areas per Zuk et al. ZIP-to-county assignment was randomized, weighted by county population, to account for overlapping geographies. The mathematical model was implemented combining coefficients of quadratic splines with the displacement metric to generate a gentrification index for each ZIP code.

2.2 Displacement metric

Prior to conducting principal component analysis (PCA), the Kaiser-Meyer-Olkin (KMO) measure and Bartlett’s Test of Sphericity were used to assess variable suitability. The KMO statistic was 0.521, indicating mediocre but acceptable sampling adequacy. Bartlett’s Test was highly significant ($p < 0.001$), confirming sufficient intercorrelation among variables to justify dimensionality reduction.

PCA was conducted on four consecutive five-year windows (starting at 2015–2019) to derive temporally grounded components without lookahead bias. Tucker congruence analysis (avg. = 0.9996; see Table A.3) confirmed high stability of the displacement metric (PC1) across these windows, supporting its validity as a consistent indicator of latent demographic turnover.

PCA Loadings Interpretations

PC1: White displacement of minority populations, capturing the primary variance in racial demographic change associated with gentrification patterns described in the literature.

38% Displacement of minorities (classic gentrification axis)

PC2: Black-specific movement, semi-independent of White/Hispanic patterns
22% Independent Black demographic shifts

PC3: Asian-specific growth/movement, largely orthogonal to others
20% Asian-specific migration dynamics

This indicates a moderately strong latent structure, with PC1 representing the principal axis of racial demographic displacement, while allowing for multi-dimensional secondary patterns. See Table A.1 for variance explained, and Table A.2 for corresponding loadings.

Among the 11.5% of PC1 outliers, only a subset with strongly positive PC1 scores are indicative of gentrification-related White influx. Negative PC1 outliers represent large demographic shifts in the opposite direction, consistent with minority-driven neighborhood change rather than classic gentrification.

To prevent extreme displacement values from overwhelming the gentrification score, I applied a soft cap: values above 5 were compressed toward the center using $5 + 0.25 \times (\text{displacement} - 5)$, and hard-capped at 10. I then divided the result by 2.3 so the displacement metric would contribute roughly 1.5 - 2 times as much explanatory weight as the other spline-based components. See Table A.4 for the full breakdown of normalized component weights.

Although the first principal component explains approximately 40% of the total variance across the demographic displacement indicators, it exhibits high stability across samples and features loadings that align with theoretical expectations (e.g., declines in minority share and increases in White non-Hispanic share). Despite the modest proportion of variance explained, PC1 represents the most coherent latent dimension of demographic turnover in the dataset and serves as a valid summary index of racial displacement pressure. This is consistent with applied use of PCA in similar multidimensional urban studies, where interpretability and empirical alignment are prioritized over maximized explained variance (Correa-Parra, Vergara-Perucich, & Aguirre-Nuñez, 2020; Australian Bureau of Statistics, 2018).

2.3 Spline Scores

To capture nonlinear temporal trends, quadratic splines were fitted to each variable - median income, higher education, homeownership, rent burden, and poverty - over the previous two years. These splines allowed the model to account for both the direction and acceleration of change, providing more nuanced representations of neighborhood dynamics than simple linear trends (Ruppert, Wand, & Carroll, 2003).

Let the temporal evolution of a exogenous variable relevant to incomplete gentri-

fication, normalized by year, be modeled as a second-degree polynomial spline:

$$f(x) = ax^2 + bx + c, \quad \text{where } x \in \{0, 1, 2\}$$

Here, $f(x)$ represents the normalized trajectory of the variable over three consecutive time points, where $x = 2$ corresponds to the current year, $x = 1$ to one year prior, and $x = 0$ to two years prior. My objective is to collapse the functional form of this equation into a single, measurable scalar quantity that summarizes the net directional change across the interval. This quantity should take on the following values:

- Zero if the trajectory is centered and symmetric (e.g., a flat or centered parabola at $x = 1$).
- Positive if the variable is increasing over time (either linearly or with upward curvature).
- Negative if the variable is decreasing or curving downward.

To this end, I propose using the average first derivative over the interval as a summary measure. The first derivative of the spline is:

$$f'(x) = 2ax + b$$

We compute the average slope over the domain $x \in \{0, 1, 2\}$ by averaging the derivative at each point:

$$\text{mean}(f') = \frac{f'(0) + f'(1) + f'(2)}{3} = \frac{(b) + (2a + b) + (4a + b)}{3} = \frac{3b + 6a}{3} = b + 2a$$

Thus, the summary score:

$$\boxed{\text{Spline Score} = b + 2a}$$

serves as a directional metric of temporal change. A positive value indicates an overall upward momentum in the variable, while a negative value indicates decline. This formulation effectively captures both linear and nonlinear trends in a single interpretable statistic.

Although a simple average rate of change, defined as the difference between endpoint values over time, offers a basic summary in terms of a linear trend, it fails to capture the full shape and dynamics of temporal evolution. In contrast, the spline-based summary score $b + 2a$, derived from a second-degree polynomial fit to a normalized variable over three consecutive years, incorporates both the linear slope and the curvature of the trend. This approach accounts for not

only the direction of change but also its acceleration or deceleration, making it particularly useful for identifying early signs of incomplete gentrification, where shifts may be non-linear or intensifying. Moreover, spline-derived scores are less sensitive to noise at individual time points, as they take advantage of the entire trajectory rather than relying solely on endpoints. This provides a more robust and informative signal, especially in cases where transient volatility or mid-period inflection would otherwise be obscured by traditional rate-of-change metrics. As such, the spline-based derivative offers a more nuanced and stable basis for comparison across spatial units and time.

To extract temporal dynamics for each ZIP code, regressions were fit to time-series data for each variable. Specifically, for each socioeconomic indicator $Y_{i,t}$, the following model was estimated:

$$Y_{i,t} = \alpha_i + \beta_{1i}f_1(t) + \beta_{2i}f_2(t) + \varepsilon_{i,t}$$

where β_{1i} and β_{2i} represent the spline slope (S_i) and curvature (C_i) coefficients, respectively, and $f_1(t)$ and $f_2(t)$ are spline basis functions in time.

The resulting spline coefficients were centered near zero with heavy tails, reflecting that most neighborhoods experienced minimal change while a smaller subset exhibited sharp socioeconomic shifts similar in structure to the displacement metric (see Figure A.1 to Figure A.6).

2.4 Incomplete Gentrification Index (IGI)

$$\begin{aligned} IGI_i = & S_i^{\text{inc}} + 2C_i^{\text{inc}} + S_i^{\text{rent}} + 2C_i^{\text{rent}} \\ & - S_i^{\text{pov}} - 2C_i^{\text{pov}} - S_i^{\text{own}} - 2C_i^{\text{own}} \\ & + S_i^{\text{edu}} + 2C_i^{\text{edu}} + D_i \end{aligned}$$

The expression above defines the Incomplete Gentrification Index (IGI_i) as a weighted summation of spline-derived slope (S) and curvature (C) terms across key socioeconomic indicators, along with the displacement metric (D_i). Variables are preserved in their natural, unstandardized state to maintain empirical interpretability and allow the resulting weights and coefficients to reflect real-world magnitudes.

To validate the IGI, I replicated the vulnerability scoring methodology from Zuk et al. (2018), focusing on Massachusetts ZIP codes. I constructed a vulnerability index using five core indicators: renter share, percentage of residents of color, percentage of households below 80% AMI (derived from income bins), share of children in poverty, and percentage of adults without a bachelor’s degree. Each variable was standardized by year using z-scores, and binary vulnerability flags

were created based on whether values exceeded 0.5 standard deviations above the mean. ZIP codes with three or more flags were retained. The final vulnerability score was calculated as the average of the five standardized indicators. This z-score-based vulnerability was then intersected with the Incomplete Gentrification Index (IGI) to identify high-vulnerability ZIP codes exhibiting partial socioeconomic turnover.

Building on the framework proposed by London, Lee, and Lipton (*The Determinants of Gentrification in the United States*), I incorporated rising housing prices—captured via spline-based income and rent trends—as indicators of emerging gentrification pressure. ZIP codes exhibiting significant upward trajectories in either income or rent splines, but lacking full demographic turnover, were flagged as likely cases of incomplete gentrification. This aligns with the original authors’ characterization of escalating housing costs as an early-stage marker of neighborhood transformation, even in the absence of full displacement or racial succession.

2.5 IGI as a signal for forecasting

Further, a LASSO regression was performed using 30 candidate predictors of future median rent to identify the most relevant variables while controlling for multicollinearity and overfitting. The gentrification index was among the variables retained after regularization, suggesting that it carries unique explanatory power beyond standard socioeconomic and demographic controls. To further assess its role, I ran a separate unpenalized linear regression, which produced a t-value of 3.42 for the index coefficient. This confirms the gentrification index as a statistically significant predictor of future rent levels at the ZIP code level.

However, rental prices are relatively sticky and respond to gentrification pressures at different time scales depending on the form of displacement, whether active or passive. A higher Incomplete Gentrification Index (IGI) may influence rental prices, but the effect can manifest unevenly across neighborhoods and with variable time lags. Due to this temporal and spatial uncertainty, incomplete gentrification cannot be treated as a clean endogenous driver of shifts in rent similar those of macroeconomic factors. Instead, it should be interpreted as a latent pressure, useful for flagging risk rather than timing market responses.

3 Conclusions

The index captures a dimension of neighborhood change often overlooked in conventional urban analysis—one that reflects structural vulnerability and incipient socioeconomic pressure before, and during, displacement. Its statistical significance across both penalized and unpenalized models underscores its relative explanatory power, even when controlling for standard economic and demographic variables. While not designed to optimize predictive performance, the index offers conceptual clarity and practical relevance, providing a scalable framework for anticipating and interpreting incomplete gentrification in real time. Displacement itself is relatively straightforward to quantify, as it can be summarized by a single principal component. Meanwhile, the stage or character of gentrification in a given ZIP code can be inferred by examining the specific spline-based components that contribute to a high IGI—revealing whether economic pressure is driven primarily by income, rent, education, or other factors.

This metric captures early-stage gentrification that traditional classifications often miss. It can be used by policymakers to flag areas at risk of displacement before it happens. Researchers can apply it to other cities or time periods for comparative analysis. Planners may integrate it into housing affordability forecasts and zoning decisions. Because it is built from public data, the method is transparent, replicable, and scalable. It is currently published on my GitHub.

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Appendix

Window	PC1	PC2	PC3
2015–2019	0.3718	0.2204	0.2027
2016–2020	0.3759	0.2238	0.2036
2017–2021	0.3773	0.2255	0.2040
2018–2022	0.3796	0.2298	0.2042

Table A.1: Explained Variance by Principal Component and Time Window

Window	PC1	PC2	PC3
2015–2019			
nonhisp_white_perc_diff	0.7211	-0.0557	-0.0204
hisp_perc_diff	-0.4829	-0.5013	-0.1284
nonhisp_black_perc_diff	-0.3553	0.8135	-0.1554
two_plus_races_perc_diff	-0.3139	-0.2886	-0.1318
nonhisp_asian_perc_diff	-0.1482	0.0236	0.9704
2016–2020			
nonhisp_white_perc_diff	0.7118	-0.0871	-0.0284
hisp_perc_diff	-0.4863	-0.4685	-0.0713
nonhisp_black_perc_diff	-0.3524	0.7787	-0.2619
two_plus_races_perc_diff	-0.3373	-0.3813	-0.0748
nonhisp_asian_perc_diff	-0.1376	0.1455	0.9591
2017–2021			
nonhisp_white_perc_diff	0.7071	-0.1033	-0.0330
hisp_perc_diff	-0.4883	-0.4582	-0.0721
nonhisp_black_perc_diff	-0.3482	0.7709	-0.2771
two_plus_races_perc_diff	-0.3514	-0.3967	-0.0450
nonhisp_asian_perc_diff	-0.1298	0.1665	0.9565
2018–2022			
nonhisp_white_perc_diff	0.7009	-0.1293	-0.0330
hisp_perc_diff	-0.4899	-0.4435	-0.0571
nonhisp_black_perc_diff	-0.3324	0.7604	-0.3152
two_plus_races_perc_diff	-0.3771	-0.4051	-0.0271
nonhisp_asian_perc_diff	-0.1266	0.2104	0.9463

Table A.2: PCA Loadings by Window and Principal Component

Window	Cosine Similarity to Mean
(2015, 2019)	0.999338
(2016, 2020)	0.999953
(2017, 2021)	0.999956
(2018, 2022)	0.999286
Global average congruence	0.999633

Table A.3: Cosine Similarity Between Windowed PCA Loadings and Global Mean

Variable	Coefficient
$z_{\text{income_spline}}$	0.1502
$z_{\text{poverty_spline}}$	-0.1706
$z_{\text{rent_spline}}$	0.1789
$z_{\text{homeownership_spline}}$	-0.1264
$z_{\text{displacement}}$	0.2517
$z_{\text{education_spline}}$	0.1102
$_{\text{CONS}}$	-0.0303

Table A.4: OLS Coefficients Explaining Gentrification Score

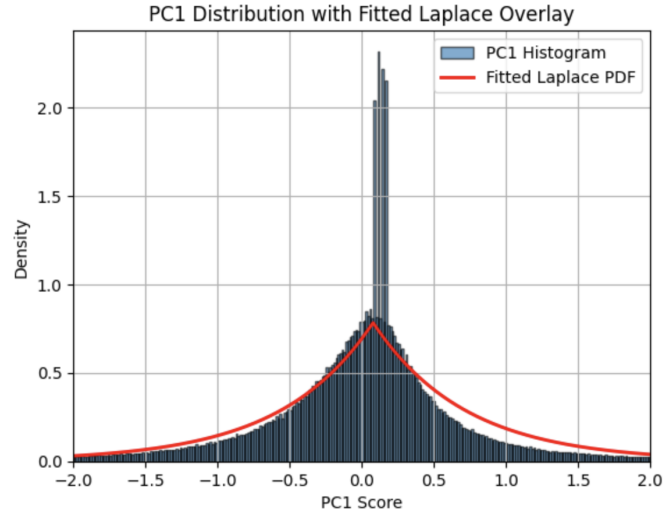


Figure A.1: Displacement metric distribution by ZIP code

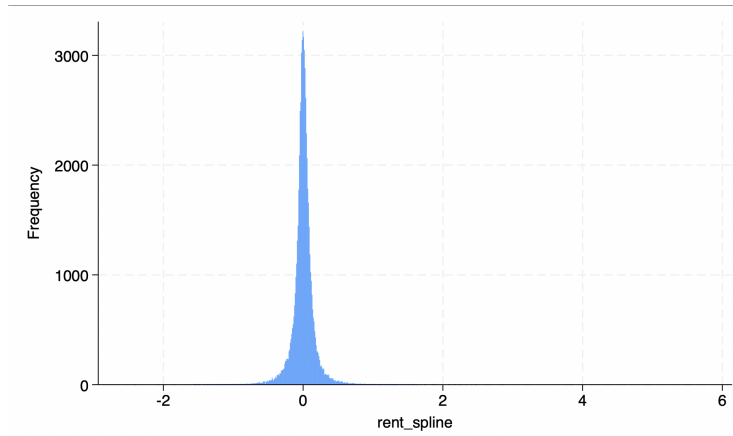


Figure A.2: Rent spline dynamics

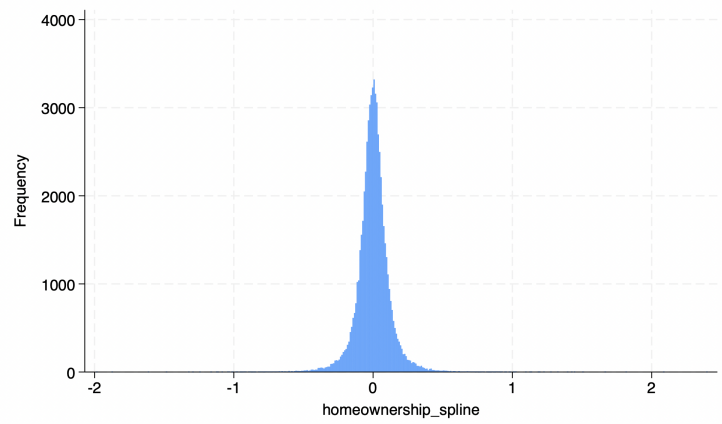


Figure A.3: Homeownership spline dynamics

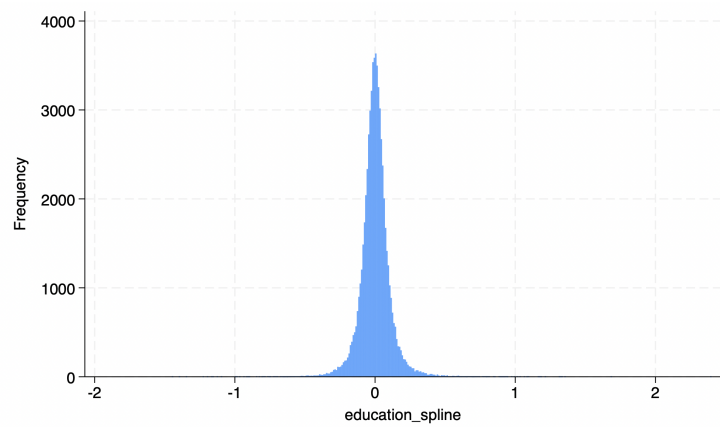


Figure A.4: Education spline dynamics

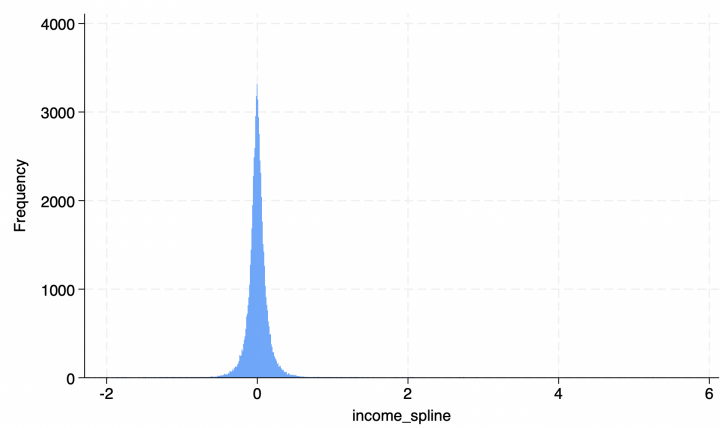


Figure A.5: Income spline dynamics

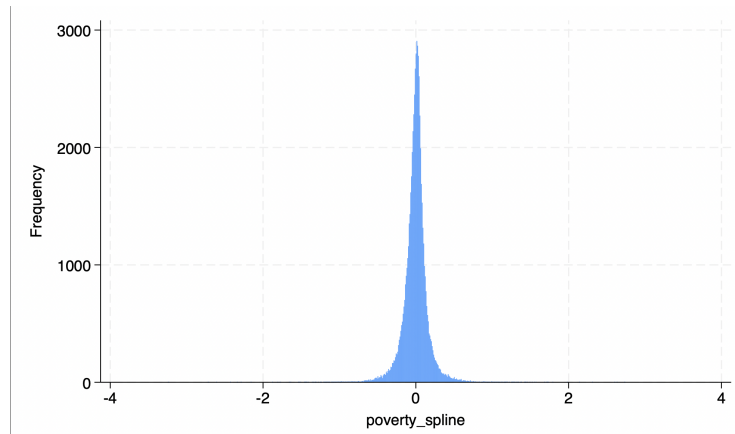


Figure A.6: Poverty spline dynamics

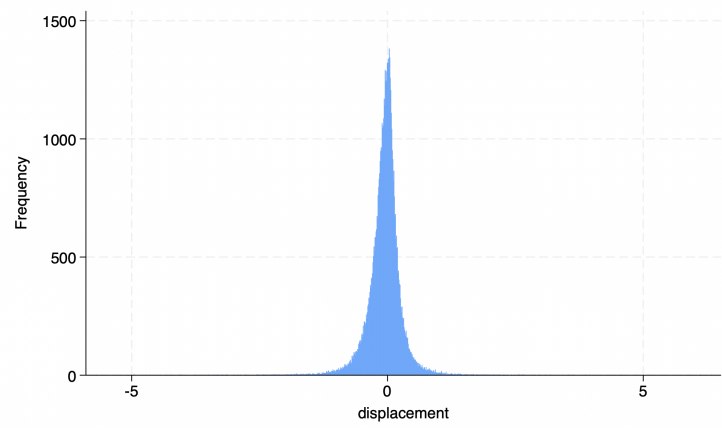


Figure A.7: Scaled displacement metric

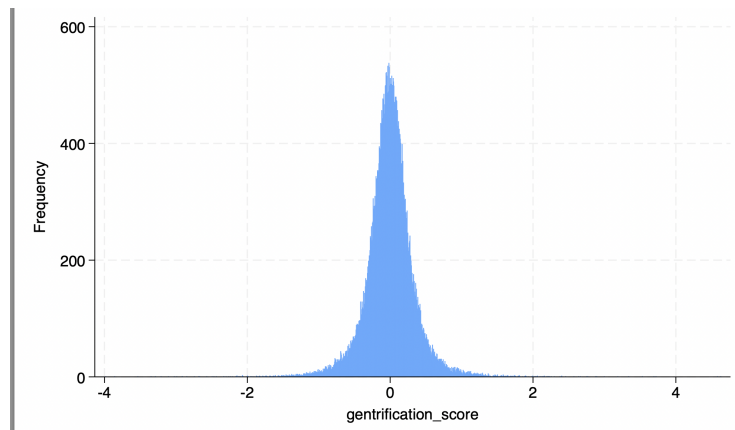


Figure A.8: Gentrification score distribution